

Notes on Symbolic Logic

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Q. What is Inductive Reasoning and Deductive Reasoning ?

Inductive Reasoning	Deductive Reasoning
Specific $\rightarrow$ General	General $\rightarrow$ Specific
We look at patterns or examples and make a likely prediction.	If the starting facts are true and the logic is correct, the conclusion must be true.
Example: • The sun has risen every morning so far. • Therefore, the sun will rise tomorrow.	Example: • All humans are mortal. • Socrates is a human. • Therefore, Socrates is mortal.

Statement and statement forms:

Definition 1.1. A *statement* is a declarative sentence that is either true or false but not both.

Example 1. The following are all statements.

- (1) $\text{\LaTeX}$ is a typesetting system for creating high-quality scientific documents.
- (2) The moon is made of cheese.
- (3) $12 \times (5 + 6) = 132$.

Example 2. The following are not statements.

- (1) Did you see the rainbow ?
- (2) $x^2 + 3x + 2 = 0$
- (3) The function $f(x)$ is not continuous at $x = 1$.

Definition 1.2. A *truth table* for a statement form P with n statements variables $P_1, P_2, \dots, P_n$ is a table with 2^n rows, consisting of columns for, at the minimum, each of the statement variables P_i and for each P . Each row of the table consists of a unique assignment of a specific truth value (true or false) to the statement variables, and the resulting truth value of P .

Definition 1.3. Given statements P and Q , the *conjunction of P and Q* is the statement denoted $P \wedge Q$, pronounced “ P and Q ”, that is true precisely when both P and Q are true and is false when at least one of P or Q is false.

Truth table of $P \wedge Q$:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Definition 1.4. Given statements P and Q , the *disjunction of P and Q* is the statement denoted $P \vee Q$, pronounced “ P or Q ”, that is true precisely when at least one of P or Q is true and is false when both P or Q are false.

Truth table of $P \vee Q$:

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Definition 1.5. Given statements P and Q , the *exclusive disjunction of P and Q* is the statement $P \veebar Q$, pronounced “ P exclusive or Q ”, that is true when exactly one of P or Q is true and is false when both P and Q are false or both P and Q are true.

Truth table of $P \vee Q$:

P	Q	$P \vee Q$
T	T	F
T	F	T
F	T	T
F	F	F

Definition 1.6. Given a statement P , the *negation of P* is the statement $\neg P$, pronounced “not P ”, that is true when P is false and is false when P is true.

Truth table of $\neg P$:

P	$\neg P$
T	F
F	T

• **Order of Precedence for logical connectives:**

$\neg \rightarrow$ highest precedence.

$\wedge, \vee$ and $\vee \rightarrow$ have same precedence, taken left-to-right.

Example 3. Interpret the following statement using order of precedence.

$$\neg P \wedge Q \vee \neg R$$

Solution: It is interpreted as :

$$((\neg P) \wedge Q) \vee (\neg R).$$

Example 4. Construct the truth table for

$$(P \vee Q) \wedge \neg (P \wedge Q).$$

Solution:

P	Q	$(P \vee Q)$	$(P \wedge Q)$	$\neg(P \wedge Q)$	$(P \vee Q) \wedge \neg(P \wedge Q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

Example 5. Construct the truth table for

$$\neg(P \vee (\neg R)) \wedge Q.$$

Solution:

P	Q	R	$\neg R$	$(P \vee (\neg R))$	$\neg(P \vee (\neg R))$	$\neg(P \vee (\neg R)) \wedge Q.$
T	T	T	F	T	F	F
T	T	F	T	F	T	T
T	F	T	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	F	T	T
F	T	F	T	T	F	F
F	F	T	F	F	T	F
F	F	F	T	T	F	F

Definition 1.7. A statement form that is true for every truth assignment to its variables is called a *tautology*. One that is false for all truth assignments to its variables is called a *contradiction*.

Example 6. For statement P , show that $P \vee (\neg P)$ is a tautology and $P \wedge (\neg P)$ is a contradiction.

Solution: We construct the truth table for $P \vee (\neg P)$ and $P \wedge (\neg P)$.

P	$\neg P$	$P \vee (\neg P)$
T	F	T
T	F	T
F	T	T
F	T	T

P	$\neg P$	$P \wedge (\neg P)$
T	F	F
T	F	F
F	T	F
F	T	F

Hence, $P \vee (\neg P)$ is a tautology and $P \wedge (\neg P)$ is a contradiction.

Definition 1.8. Two statement forms with the same variables are called *logically equivalent*, with equivalence denoted via the symbol $\equiv$, if the statement forms have the same truth value for every possible truth assignment to the statement variables of the statement form.

Example 7. The statements $P \vee Q$ and $(P \vee Q) \wedge \neg(P \wedge Q)$ are equivalent, as verified by the truth tables in Definition 1.5 and in Example 4. We write

$$P \vee Q \equiv (P \vee Q) \wedge \neg(P \wedge Q)$$

Theorem 1.1. Let P , Q and R be statements, t a tautology and c a contradiction. The following logical equivalences, called *logical equivalence laws*, hold.

- | | |
|--------------------------------|---|
| 1. <i>Commutative</i> | $P \wedge Q \equiv Q \wedge P$ $P \vee Q \equiv Q \vee P$ |
| 2. <i>Associative</i> | $P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$ $P \vee (Q \vee R) \equiv (P \vee Q) \vee R$ |
| 3. <i>Distributive</i> | $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$ $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$ |
| 4. <i>Identity</i> | $P \wedge t \equiv P$ $P \vee c \equiv P$ |
| 5. <i>Negation</i> | $P \vee \neg P \equiv t$ $P \wedge \neg P \equiv c$ |
| 6. <i>Double Negative</i> | $\neg(\neg P) \equiv P$ |
| 7. <i>Idempotent</i> | $P \wedge P \equiv P$ $P \vee P \equiv P$ |
| 8. <i>Universal Complement</i> | $\neg t \equiv c$ $\neg c \equiv t$ |

Theorem 1.2. (De Morgan's Laws) For statements P and Q , we have

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

and

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

Proof : (i) $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$

Truth table :

P	Q	$\neg P$	$\neg Q$	$(P \wedge Q)$	$\neg(P \wedge Q)$	$\neg P \vee \neg Q$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

(ii) $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

Truth table :

P	Q	$\neg P$	$\neg Q$	$(P \vee Q)$	$\neg(P \vee Q)$	$\neg P \wedge \neg Q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Exercises

- Construct a truth table for each of the following. Assume P , Q and R are statements.

(a) $\neg(P \vee \neg Q) \vee \neg(P \wedge Q)$

(b) $[(P \wedge \neg Q) \vee R] \vee (\neg R \vee Q)$

(c) $P \vee (\neg P)$

- Show the following logical equivalence hold. Assume that P , Q and R are statements.

(a) $P \vee Q \equiv \neg(\neg P \wedge \neg Q)$

$$(b) P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

3. Determine if each of the following pairs of statements are logically equivalent or not. Assume all variables represent statements.

$$(a) P \vee Q \quad , \quad (P \vee Q) \vee (\neg P \wedge \neg Q)$$

$$(b) \neg(\neg P) \vee \neg P \quad , \quad Q \vee \neg Q$$

$$(c) P \vee (P \wedge \neg Q) \quad , \quad P \wedge \neg Q$$

Conditional and Biconditional connective

Definition 1.9. A *conditional statement* (or *implication*) is one of the form “if P , then Q ”, where P and Q are statements, and is denoted $P \Rightarrow Q$. The statement P is called the *hypotheses* (or *assumption*, *premise*, or *antecedent*) of the conditional statement. The statement Q is called the *conclusion* (or *consequence*) of the implication. The conditional statement $P \Rightarrow Q$ is false only when P is true and Q is false.

Truth table of $P \Rightarrow Q$:

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Theorem 1.3. Let P and Q be statements. Then,

$$P \Rightarrow Q \equiv \neg P \vee Q$$

Proof : Let P and Q be statements. Then,

P	Q	$P \Rightarrow Q$	$\neg P$	$\neg P \vee Q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Because $P \Rightarrow Q$ and $\neg P \vee Q$ have the same truth value for all truth assignments for P and Q , they are logically equivalent.

Example 8. Show that $P \Rightarrow (Q \Rightarrow R) \not\equiv (P \Rightarrow Q) \Rightarrow R$ using truth table.

Definition 1.10. The *contrapositive* to the statement $P \Rightarrow Q$ is the statement $\neg Q \Rightarrow \neg P$. The *converse* to $P \Rightarrow Q$ is the statement $Q \Rightarrow P$, and the *inverse* statement to $P \Rightarrow Q$ is $\neg P \Rightarrow \neg Q$.

Theorem 1.4. A conditional statement is logically equivalent to its contrapositive but not to either its converse or inverse.

Proof : Truth table

P	Q	$\neg P$	$\neg Q$	$P \Rightarrow Q$	$\neg Q \Rightarrow \neg P$	$\neg P \Rightarrow \neg Q$	$Q \Rightarrow P$
T	T	F	F	T	T	T	T
T	F	F	T	F	F	T	T
F	T	T	F	T	T	F	F
F	F	T	T	T	T	T	T

Note: The *converse* and the *inverse* of an implication are *logically equivalent*.

Definition 1.11. The *negation* of the conditional statement $P \Rightarrow Q$ is $P \wedge \neg Q$.

Truth table :

P	Q	$\neg Q$	$P \Rightarrow Q$	$P \wedge \neg Q$
T	T	F	T	F
T	F	T	F	T
F	T	T	T	F
F	F	T	T	F

Definition 1.12. A *biconditional statement* is one of the form “ P if and only if Q ”, where P and Q are statements, and is denoted $P \Leftrightarrow Q$. It is true when P and Q have the same truth value and false when their truth values are different.

Truth table of $P \Leftrightarrow Q$:

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Example 9. Build the truth table for

$$P \Leftrightarrow (Q \Rightarrow \neg R)$$

Solution : Truth table

P	Q	R	$\neg R$	$Q \Rightarrow \neg R$	$P \Leftrightarrow (Q \Rightarrow \neg R)$
T	T	T	F	F	F
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	T	T	T
F	T	T	F	F	T
F	T	F	T	T	F
F	F	T	F	T	F
F	F	F	T	T	F

Theorem 1.5. For statements P and Q , $P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$.

Exercises

- Create a truth table for each of the following. Assume P , Q and R are statements.
 - $(\neg P \vee Q) \Rightarrow (\neg Q \Rightarrow P)$
 - $(P \vee Q) \Rightarrow (R \wedge Q)$
 - $(P \Leftrightarrow R) \wedge (P \Leftrightarrow Q)$
 - $P \Leftrightarrow (Q \vee (R \Rightarrow P))$
- Determine, if each of the following is a tautology, contradiction, or neither. Assume all variables represent statements.
 - $(Q \Leftrightarrow P) \Leftrightarrow (P \Leftrightarrow R)$
 - $\neg(Q \Rightarrow P) \Rightarrow P$

$$(c) [P \wedge (Q \Leftrightarrow R)] \Rightarrow \neg R$$

$$(d) [P \Rightarrow (Q \Rightarrow P)] \vee (Q \Leftrightarrow R)$$

3. Suppose P , Q , R , S and T are statements. Show that

$$P \Rightarrow (Q \Rightarrow (R \Rightarrow (S \Rightarrow T)))$$

and

$$P \Rightarrow (Q \Rightarrow (R \Rightarrow (S \Rightarrow \neg T)))$$

are not logically equivalent.

Arguments

Definition 1.13. An *argument* is a finite list of statements $P_1, P_2, \dots, P_{n-1}$, called the *assumptions* (or *premises* or *hypotheses*) of the argument, and a single statement P_n , called the *conclusion* of the argument. We write such an argument in the following way. The symbol $\therefore$ is read as “therefore”.

$$\begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_{n-1} \\ \hline \therefore P_n \end{array}$$

Definition 1.14. An argument is called *valid* if whenever every assumption of the argument is true, then the conclusion is also true. If there is a case when the assumptions are true and conclusion is false, then the argument is called *invalid*.

Example 10. Show that the following argument is invalid.

$$\begin{array}{c} L \Rightarrow C \\ C \\ \hline \therefore L \end{array}$$

Solution : We built a truth table with a column for every assumption and the conclusion.

L	C	$L \Rightarrow C$
T	T	T
T	F	F
F	T	T
F	F	T

The third row, as indicated by the arrow, shows that this argument is invalid. The assumptions are both true but the conclusion is false.

Example 11. Show the argument below is valid.

$$\begin{array}{l}
 A \Rightarrow C \\
 B \Leftrightarrow C \\
 \neg A \\
 \hline
 \therefore A \Rightarrow B
 \end{array}$$

Solution : We built a truth table with a column for each assumption and a column for the conclusion.

A	B	C	$A \Rightarrow C$	$B \Leftrightarrow C$	$\neg A$	$A \Rightarrow B$
T	T	T	T	T	F	T
T	T	F	F	F	F	T
T	F	T	T	F	F	F
T	F	F	F	T	F	F
F	T	T	T	T	T	T
F	T	F	T	F	T	T
F	F	T	T	F	T	T
F	F	F	T	T	T	T

Here, conclusion, $A \Rightarrow B$, is true. Thus, the argument is valid.

Remark :

1. Using a truth table to show an argument is invalid requires finding a *single* row that satisfies certain conditions (every assumptions is true and the conclusion is false). Showing an argument is valid, we have to consider *every row* where all the assumptions are true and eventually the conclusion is also true.
2. It is important to note that by claiming an *argument* is valid, we are making a claim only about the argument rather than the individual

statements within the argument. This is highlighted by the following two examples, both of which contain valid arguments.

Example 12. Show the following argument is valid.

The moon is made of cheese.
 If the moon is made of cheese, then the world is flat.

 Therefore, the world is flat.

Example 13. Investigate the validity of the following argument.

$$\frac{P \wedge \neg P}{\therefore Q}$$

Solution : Consider the following truth table:

P	Q	$\neg P$	$P \wedge \neg P$
T	T	F	F
T	F	F	F
F	T	T	F
F	F	T	F

Notice that there are no rows where the hypothesis $P \wedge \neg P$ is true. In order to say that an argument is *invalid*, there must be a row of the truth table in which all the hypotheses are true and the conclusion is false. Since this does not exist, we say that the argument is valid *vacuously*.

Definition 1.15. A *rule of inference* is an argument that is valid.

- Following are some of the common named arguments :

Modus Ponens

$$\frac{P \Rightarrow Q \quad P}{\therefore Q}$$

Transitivity

$$\frac{P \Rightarrow Q \quad Q \Rightarrow R}{\therefore P \Rightarrow R}$$

Generalization

$$\frac{P(\text{or } Q)}{\therefore P \vee Q}$$

Modus Tollens

$$\frac{P \Rightarrow Q \quad \neg Q}{\therefore \neg P}$$

Conjunction

$$\frac{P \quad Q}{\therefore P \wedge Q}$$

Specialization

$$\frac{P \wedge Q}{\therefore P(\text{or } Q)}$$

$$\begin{array}{l}
 \textbf{Elimination} \\
 P \vee Q \\
 \neg P (\text{or } \neg Q) \\
 \hline
 \therefore Q (\text{or } P)
 \end{array}$$

$$\begin{array}{l}
 \textbf{Cases} \\
 P \vee Q \\
 P \Rightarrow R \\
 Q \Rightarrow R \\
 \hline
 \therefore R
 \end{array}$$

Example 14. Show that Modus Tollens is a valid argument.

Solution : Truth table

P	Q	$P \Rightarrow Q$	$\neg Q$	$\neg P$
T	T	T	F	F
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

The last row of the truth table is the only one in which both hypotheses, $P \Rightarrow Q$ and $\neg Q$, are true. Note that the conclusion, $\neg P$, is true as well, showing that the argument is indeed valid.

- Following two are the most-common invalid arguments :

$$\begin{array}{l}
 \textbf{Converse Error} \\
 P \Rightarrow Q \\
 Q \\
 \hline
 \therefore P
 \end{array}$$

$$\begin{array}{l}
 \textbf{Inverse Error} \\
 P \Rightarrow Q \\
 \neg P \\
 \hline
 \therefore \neg Q
 \end{array}$$

Exercises

1. Determine if the following arguments are valid or invalid.

$$\begin{array}{l}
 (a) \quad \begin{array}{l} P \vee Q \\ P \Rightarrow Q \\ \hline \therefore Q \end{array}
 \end{array}$$

$$\begin{array}{l}
 (b) \quad \begin{array}{l} P \vee Q \vee R \\ \neg Q \vee \neg R \\ \hline \therefore P \end{array}
 \end{array}$$

$$\begin{array}{l}
 (c) \quad \begin{array}{l} P \vee Q \\ P \Rightarrow Q \\ \hline \therefore Q \end{array}
 \end{array}$$

$$\begin{array}{l}
 (d) \quad \begin{array}{l} P \vee Q \vee R \\ \neg Q \vee \neg R \\ \hline \therefore P \end{array}
 \end{array}$$

2. Explain why the following theorem is true.

Theorem : The argument below is valid if and only if
 $(P_1 \wedge P_2 \wedge \dots P_{n-1}) \Rightarrow P_n$ is a tautology.

$$\begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_{n-1} \\ \hline \therefore P_n \end{array}$$

3. Check whether the following statement is valid or invalid.

$$\begin{array}{c} P \\ Q \\ R \\ \hline \therefore P \vee \neg P \end{array}$$

Logical Deduction

Definition 1.16. A *logical deduction* is a process for showing an argument is valid. It is a finite sequence of statements, where every term of the sequence is either a hypothesis of the argument or it follows from a single rule of inference or logical equivalence applied to previous terms of the sequence, and the final term of the sequence is the conclusion of the argument. Each statement is accompanied by a justification for why it is allowed to appear in the deduction.

Example 15. Show that the following arguments, where P , Q and R are statements is valid.

$$\begin{array}{c} P \\ P \Rightarrow Q \\ Q \Rightarrow R \\ \hline \therefore R \end{array}$$

Solution : Our logical deduction is as follows :

- | | |
|----------------------|-------------------|
| 1. P | Assumption |
| 2. $P \Rightarrow Q$ | Assumption |
| 3. $Q \Rightarrow R$ | Assumption |
| 4. Q | Modus Ponens 1, 2 |
| 5. R | Modus Ponens 3, 4 |

Example 16. Use a logical deduction to show that the following argument, where P , Q and R are statements, is valid.

$$\begin{array}{l} P \\ P \Rightarrow Q \\ \neg Q \vee R \\ \hline \therefore R \end{array}$$

Solution : Our deduction is :

1. P	Assumption
2. $P \Rightarrow Q$	Assumption
3. $\neg Q \vee R$	Assumption
4. Q	Modus Ponens 1, 2
5. $\neg(\neg Q)$	Double Negative 4
6. R	Elimination 3, 5

Example 17. Prove that the following argument, assuming all variables represent statements, is valid.

$$\begin{array}{l} (\neg P \vee Q) \Leftrightarrow R \\ S \vee \neg Q \\ \neg U \\ P \Rightarrow U \\ (\neg P \wedge R) \Rightarrow \neg S \\ \hline \therefore \neg Q \end{array}$$

Solution : Our deduction is :

1. $(\neg P \vee Q) \Leftrightarrow R$	Assumption
2. $S \vee \neg Q$	Assumption
3. $\neg U$	Assumption
4. $P \Rightarrow U$	Assumption
5. $(\neg P \wedge R) \Rightarrow \neg S$	Assumption
6. $\neg P$	Modus Tollens 3, 4
7. $((\neg P \vee Q) \rightarrow R) \wedge (R \Rightarrow (\neg P \vee Q))$	Theorem 1.5 1
8. $(\neg P \vee Q) \Rightarrow R$	Specialization 7
9. $\neg P \vee Q$	Generalization 6
10. R	Modus Ponens 8, 9
11. $\neg P \wedge R$	Conjunction 6, 10
12. $\neg S$	Modus Ponens 5, 11
13. $\neg Q$	Elimination 2, 12

Exercises

1. Use a logical deduction to show that the following are valid arguments. Assume all variables represent statements.

$$(a) \begin{array}{l} P \Rightarrow Q \\ Q \Rightarrow R \\ \neg R \end{array}$$

$$\hline \therefore \neg P$$

$$(c) \begin{array}{l} P \wedge (Q \wedge R) \\ (Q \wedge R) \Rightarrow S \end{array}$$

$$\hline \therefore S$$

$$(e) \begin{array}{l} \neg P \Leftrightarrow (S \vee Q) \\ S \vee (Q \wedge T) \\ W \Rightarrow P \end{array}$$

$$\hline \therefore \neg W$$

$$(b) \begin{array}{l} Q \wedge P \\ (P \wedge Q) \Rightarrow \neg R \\ P \vee S \end{array}$$

$$\hline \therefore S$$

$$(d) \begin{array}{l} (P \vee Q) \vee R \\ \neg Q \wedge \neg R \end{array}$$

$$\hline \therefore P$$

$$(f) \begin{array}{l} \neg P \neg \neg Q \\ \neg T \Leftrightarrow (P \wedge Q) \\ \neg T \vee S \\ S \Rightarrow (WU) \end{array}$$

$$\hline \therefore U$$

2. Determine the errors in the following logical deductions. Assume all variables represent statements.

- | | | |
|-----|---------------------------|--------------------|
| | 1. $P \wedge Q$ | Assumption |
| (a) | 2. $P \Rightarrow R$ | Assumption |
| | 3. R | Modus Ponens 1, 2 |
| | 1. $P \vee Q$ | Assumption |
| | 2. $P \Rightarrow R$ | Assumption |
| (b) | 3. $\neg Q$ | Assumption |
| | 4. R | Modus Ponens 1, 2 |
| | 1. $(P \vee Q) \wedge R$ | Assumption |
| | 2. $P \Rightarrow \neg R$ | Assumption |
| | 3. R | Specialization 1 |
| (c) | 4. $P \vee Q$ | Generalization 1 |
| | 5. $\neg P$ | Modus Tollens 2, 3 |
| | 6. Q | Elimination 4, 5 |
| | 1. $(P \vee Q) \wedge R$ | Assumption |
| | 2. $P \Rightarrow \neg R$ | Assumption |
| | 3. R | Specialization 1 |
| (d) | 4. $P \vee Q$ | Generalization 1 |
| | 5. $\neg P$ | Modus Tollens 2, 3 |
| | 6. Q | Elimination 4, 5 |